

can handle low to medium loads, AC and DC, resistive and inductive loads like solenoid valves, fans, and heaters, but not large inrush currents.

The PT570730 relay is a miniature industrial relay for general-purpose applications. It has silver contacts and is used in control and automation applications. Other applications of this relay include consumer electronics, and automotive and power management applications.

Silver nickel (AgNi). It is often used in the context of resistive load and current stability. The modest amount of nickel (10%) is mainly used to mechanically harden the silver and boost the resistance of the contact, making it more robust under higher electrical loads. It is appropriate for resistive loads at the contact's full nominal current rating as well as other loads with lower load currents.

Silver nickel is a cost-effective and high-performing, general-purpose material that is frequently used as the standard for many power relays. When silver nickel contacts are gold plated, they are suitable for dry loads, measuring and switching circuits.

The Finder 60 series general-purpose relay is a good example of relays with AgNi contacts. It has gold-plated AgNi contacts and is suitable for industrial and shipyard applications, and road and tunnel lighting. It offers a lockable test button with a mechanical flag indicator.

Silver tin oxide (AgSnO₂). AgSnO₂ is used in applications that will produce surge current, such as applications with a capacitive load, inductive load, motor load, etc. It is well suited for higher loads, particularly for high inrush currents with short rise times. Silver tin oxide is widely used to replace the more hazardous silver cadmium oxide as a relay contact material.

Important considerations for relay selection

- Environmental adaptability requirements
- Input and output parameters
- Time parameters
- Contact life
- Dimensions
- Weight
- Installation size
- Installation method
- Sealing

Hongfa relay HF46F-G series is a great example of a relay that introduces a difference in its contact material in different applications. The relay contact material given in the specification is divided into two materials: AgSnO₂ and AgNi. That is, the contact material of HF46F-G/XTT (with T) is AgSnO₂, while the contact material of HF46F-G/XX (without T) is AgNi.

Silver cadmium oxide (AgCdO). Silver cadmium oxide contacts have long been utilised to switch high-energy, arc-producing loads. They are less electrically conductive than fine silver contacts, but they have better material transfer and arcing resistance. They do have a somewhat higher contact assembly heat and a higher interface resistance between mated contacts.

An example of a relay that uses AgCdO contacts would be the T92S7A12-240 general-purpose relay. It is a high-power relay with a wash-tight, tape-sealed plastic box enclosure. The T92-series is designed to control compressor loads and has a printed circuit board mount, printed circuit board terminal, non-latching coil, and 240V AC coil voltage. It can be used in HVAC applications.

Silver tin indium oxide. Although silver tin indium oxide connections are more difficult to come by, they are more resistant to arc erosion and welding than silver cadmium oxide contacts. They are,



Fig. 8: T92S7A12-240 general-purpose relay with AgCdO contacts (Credit: element14)



Fig. 9: T9AS1D14-24 relay with silver tin indium oxide contacts (Credit: element14)

however, less electrically conductive and more difficult to work with than silver cadmium oxide connections. Also, since they are more expensive than silver cadmium oxide at the moment, many relay users restrict their use to incandescent lamp loads and capacitors with high inrush current during contact bounce.

The T9AS1D14-24 relay, which is well suited to power management and HVAC applications, has silver tin indium oxide contacts. It has an option for load connections via 6.35mm terminals. **EFY**

This writeup is inspired by an article authored by the content team at Mornsun. The co-author, Aaryaa Padhyegurjar, is an Industry 4.0 enthusiast at EFY with a keen interest in innovation and research.